

Jehyeok Yeon

Ph.D. Candidate at Max Planck Institute for Intelligent Systems

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Research Interests

My research develops frameworks for evaluating frontier agents on **non-verifiable tasks**, such as long-horizon automated AI R&D, where models increasingly exceed human competence and even exploit evaluation loopholes to artificially improve measured performance. I work on **calibrating estimates of model capability** against the **reliability of oversight**, so that evaluations can remain meaningful as capabilities rise and deployment distributions shift. My current research focuses include:

- **Evaluating non-verifiable tasks to locate frontier bottlenecks:** Methods for measuring frontier agent performance on realistic, non-verifiable tasks with no clean pass/fail, built so results point at actionable capability bottlenecks grounded in deployment task distributions rather than toy settings.
- **Creating a science of oversight capabilities:** Mapping exactly when, where, and how human and automated oversight break down as capability rises so that trust in AI capabilities is properly calibrated with robust estimates of oversight reliability.

Education

Ph.D. in Computer Science

July 2026 – October 2029

Max Planck Institute for Intelligent Systems

Advisor: Maksym Andriushchenko

BS in Computer Science + Linguistics

August 2022 – December 2025

University of Illinois Urbana-Champaign (UIUC)

GPA 3.92/4.0

Advisor: Prof. Gagandeep Singh

Selected Publications

- **Jehyeok Yeon**, Federico Cinus, Yifan Wu, Luca Luceri. *GSAE: Graph-Regularized Sparse Autoencoders for Robust LLM Safety Steering*. [NeurIPS 2026](#); [ICML 2026 AI4GOOD](#)
- **Jehyeok Yeon**, Hyeonjeong Ha, Qiusi Zhan, Heng Ji. *Securing Multimodal AI through Internal Information Decomposition*. [ICML 2026 \(Spotlight\)](#)
- **Jehyeok Yeon**, Ben Rank, Maksym Andriushchenko. *InferenceBench: A Benchmark for Open-Ended LLM Inference Optimization by AI Agents*. [ICML 2026 AIWILD \(Spotlight\)](#)
- **Jehyeok Yeon**, Isha Chaudhary, Gagandeep Singh. *Certifying Robustness of Agent Tool-Selection Under Adversarial Attacks*. [ICLR 2026 AIWILD](#)

- Hangoo Kang*, **Jehyeok Yeon***, Gagandeep Singh. *TRAP: Targeted Redirecting of Agentic Preferences*. **NeurIPS 2025** (*indicates equal contribution)
- Ben Rank*, Lena Libon*, **Jehyeok Yeon**, Jeremy Qin, David Schmotz, Derck Prinzhorn, Daniel Donnelly, Maksym Andriushchenko. *ResearchArena: Evaluating Sabotage and Monitoring in Automated AI R&D*. **COLM 2026 WAB** (*indicates equal contribution)

Research Experience

Anthropic

November 2026 – February 2027

Research Fellow with [TBD]

London AI Safety Research Labs (LASR Labs)

July 2026 – October 2026

Research Scholar with Justin Olive and Magda Dubois

- Conducting Scalable Oversight research with UK AI Security Institute and Generality Labs.

Max Planck Institute for Intelligent Systems (MPI-IS)

January 2026 – June 2026

Visiting Student Researcher with Maksym Andriushchenko

- Conducted Agentic R&D research to create long-horizon benchmarks for AI optimization tasks.

University of Southern California (USC)

May 2025 – June 2026

Visiting Student Researcher with Prof. Luca Luceri

- Conducted Information Geometry research to model and analyze LLM internals as complex, non-linear data structures.

University of Illinois Urbana-Champaign (UIUC)

August 2024 – June 2026

Research Assistant with Prof. Gagandeep Singh and Prof. Heng Ji

- Conducted quantitative certification research to model and evaluate the safety of agentic AI systems.
- Conducted AI Security research to develop a novel indirect prompt injection method utilizing semantic injection on images.

Work Experience

Intradiem

May 2025 – August 2025

Machine Learning Intern

- Reduced false-negative burnout classifications by 35% by building feature extraction pipelines with PyTorch, Temporal, and distributed SQL.
- Built predictive models to detect burnout anomalies across terabytes of call center telemetry data.

Hanwha Life Insurance

June 2024 – August 2024

AI Data Scientist

- Deployed a RAG Agent utilizing LangChain, ChromaDB, and fine-tuned instruction models.
- Optimized hybrid retrieval and prompt orchestration to improve average query accuracy by 40% while maintaining resolution overhead in domain-specific QA tasks.

ATLAS

May 2023 – August 2024

Machine Learning Intern

- Engineered a real-time predictive model utilizing hyper-local streaming sensor data for the city of Chicago, enhancing the city’s resource allocation and optimizing response times by 15% during peak events.
- Minimized inference latency for meteorological forecasting using TensorFlow Lite and ElasticSearch.

Honors and Awards

- NSF Graduate Research Fellowship Program (GRFP) (2026)
- AI4GOOD Travel Grant (2026)
- Sandia Data Challenge Finalist Award (2025)
- UIUC Get Experience Scholarship (2024)

Talks

What Open-Ended Benchmarks Reveal About AI Research Automation

- Google DeepMind; Turing; CAMEL AI; KAIST University (2026)

Rise of Indirect Prompt Injection Methods on Open Environment Agents

- AI Safety Tübingen Reading Group (2026)

Academic Service

Reviewer

- NeurIPS 2026, PALM Workshop @ NeurIPS 2026, AI4GOOD Workshop @ ICML 2026, Agents in the Wild @ ICML 2026

Teaching Assistant

- AI Safety Course Summer 2026 @ University of Tübingen

Skills

- Languages: Python, Java, C++, SQL
- Frameworks: PyTorch, TensorFlow, Transformers, vLLM, TransformerLens, LangChain, ChromaDB, W&B, Docker, Git, Temporal, MLflow, ElasticSearch, Spring Boot

Publications

- **Jehyeok Yeon**, Hyeonjeong Ha, Qiusi Zhan, Heng Ji. *Securing Multimodal AI through Internal Information Decomposition*. [ICML 2026 \(Spotlight\)](#).
- **Jehyeok Yeon**, Ben Rank, Maksym Andriushchenko. *InferenceBench: A Benchmark for Open-Ended LLM Inference Optimization by AI Agents*. [ICML 2026 AIWILD \(Spotlight\)](#)
- **Jehyeok Yeon**, Isha Chaudhary, Gagandeep Singh. *Certifying Robustness of Agent Tool-Selection Under Adversarial Attacks*. [ICLR 2026 AIWILD](#)
- **Jehyeok Yeon**, Federico Cinus, Yifan Wu, Luca Luceri. *GSAE: Graph-Regularized Sparse Autoencoders for Robust LLM Safety Steering*. [NeurIPS 2026; ICML 2026 AI4GOOD](#)
- Hangoo Kang*, **Jehyeok Yeon***, Gagandeep Singh. *TRAP: Targeted Redirecting of Agentic Preferences*. [NeurIPS 2025](#). (*indicates equal contribution)
- **Jehyeok Yeon**, Lawrence Angrave. *The Power of Friendship: Analyzing Leadership and Adversarial Attacks in Multi-Agent Collaboration*. [ACM Collective Intelligence 2025 Poster Acceptance](#).
- Ben Rank*, Lena Libon*, **Jehyeok Yeon**, Jeremy Qin, David Schmotz, Derck Prinzhorn, Daniel Donnelly, Maksym Andriushchenko. *ResearchArena: Evaluating Sabotage and Monitoring in Automated AI R&D*. [COLM 2026 WAB](#) (*indicates equal contribution)